

Nimbus Alerting Service

Operations Handbook, version 3.2 (fictional document for testing)

1. OVERVIEW

1.1 Purpose

Nimbus is an internal alert routing service. It receives monitoring events from many systems, removes noise, decides who must be told, and delivers notifications by phone, chat or email. Its main goal is to make sure the right engineer hears about a serious problem quickly, without being flooded by repeated messages about the same fault.

1.2 Architecture

Nimbus has four components. The ingest gateway accepts events over TLS on port 9443. A stream bus buffers the events. The rule engine evaluates each event against the configured rules. The notifier delivers messages to the on-call responders. The metrics endpoint runs on port 9102 and the administration console runs on port 8088.

2. DEPLOYMENT

2.1 Requirements

Each Nimbus node needs at least 4 vCPUs and 8 GB of RAM. A production cluster should run three nodes so that one can fail without interrupting alert delivery. Clocks must be synchronised, because deduplication and escalation depend on accurate timestamps.

2.2 Configuration

Nimbus is configured through environment variables. The most important settings are listed below.

Variable	Default	Meaning
NIMBUS_RETENTION_DAYS	45	How long processed alerts are kept before deletion.
NIMBUS_MAX_BATCH	250	Maximum events the rule engine processes in one batch.
NIMBUS_DEDUP_WINDOW_MIN	15	Length of the deduplication window in minutes.
NIMBUS_NOTIFIER_WORKERS	4	Number of parallel notification workers.

2.3 Ports and endpoints

Service	Port	Protocol
Ingest gateway	9443	HTTPS (TLS)
Metrics	9102	HTTP, Prometheus format
Administration console	8088	HTTPS

3. ALERT PROCESSING

3.1 Deduplication

When many systems report the same fault, Nimbus groups them. For each event it computes a fingerprint from the source type, the rule name and the severity, using a SHA-256 hash. If another event with the same fingerprint arrives within the deduplication window, which is 15 minutes by default, Nimbus does not send a new notification. It increases a counter on the existing alert instead. This is why forty failing servers with the same problem produce one page, not forty.

3.2 Severity levels

Severity	Expected response	Delivery
SEV1	A responder must acknowledge within 2 minutes	Phone call and chat
SEV2	Acknowledge within 30 minutes	Chat and email
SEV3	Handle by the next business day	Email only

3.3 Escalation

If a SEV1 alert is not acknowledged after 10 minutes, Nimbus escalates it to the secondary on-call engineer. If it is still unacknowledged after 25 minutes, Nimbus escalates to the engineering manager. Each escalation step is recorded in the alert history. Acknowledging an alert stops further escalation but does not close the alert.

3.4 Maintenance windows

During planned work, a maintenance window suppresses notifications for the affected systems. A window can last at most 8 hours and requires an approval token issued by an administrator. Events received during a window are still stored, so they can be reviewed afterwards.

4. ERRORS AND TROUBLESHOOTING

4.1 Error codes

Code	Meaning	Typical action
ERR-4107	A rule contains invalid syntax and was rejected.	Fix the rule definition and reload.
ERR-5203	The notifier timed out after 12 seconds while contacting a delivery provider.	Check provider status and retry.
ERR-3310	The stream bus backlog exceeds 50,000 messages.	Follow the backlog recovery procedure.
ERR-2001	An event was rejected because its authentication token expired.	Request a new token.

4.2 Backlog recovery

When ERR-3310 appears, first confirm that the rule engine is running. Then raise NIMBUS_NOTIFIER_WORKERS from 4 to 6 so that more notifications are delivered in parallel. Pause all SEV3 rules until the backlog falls below 10,000 messages, and resume them one at a

time. Do not restart the stream bus while it is draining, because unprocessed events could be lost.

5. SECURITY AND PRIVACY

5.1 Access control

Nimbus has three roles. Viewers can read alerts. Responders can acknowledge and escalate alerts. Administrators can change rules, issue approval tokens and edit configuration. Access tokens expire after 24 hours and must then be renewed.

5.2 Data handling

Alert payloads are stored encrypted at rest. On-call phone numbers are never written to application logs. Processed alerts are deleted automatically once they are older than the retention period.

5.3 Test contact

For the staging environment only, the test escalation contact is Jane Roe, email jane.roe@example.com, phone +1-555-0100. This contact is fictional and used only to verify that notifications are delivered.

6. FREQUENTLY ASKED QUESTIONS

6.1 Why did I receive one notification for many failing hosts?

Nimbus groups repeated events that share the same fingerprint and sends a single notification for the group. The alert shows how many events were folded into it.

6.2 What happens if nobody responds to a critical alert?

Critical alerts are passed up the chain automatically: first to the secondary on-call engineer, then to the engineering manager, until someone acknowledges.

6.3 Can alerts be silenced for planned work?

Yes. Use a maintenance window, which needs an approval token and cannot be longer than 8 hours.